

Out of the Loop: Enhancing Documentation and Transparency in Collaborative Causal Loop Diagrams to Capture Multiple Perspectives

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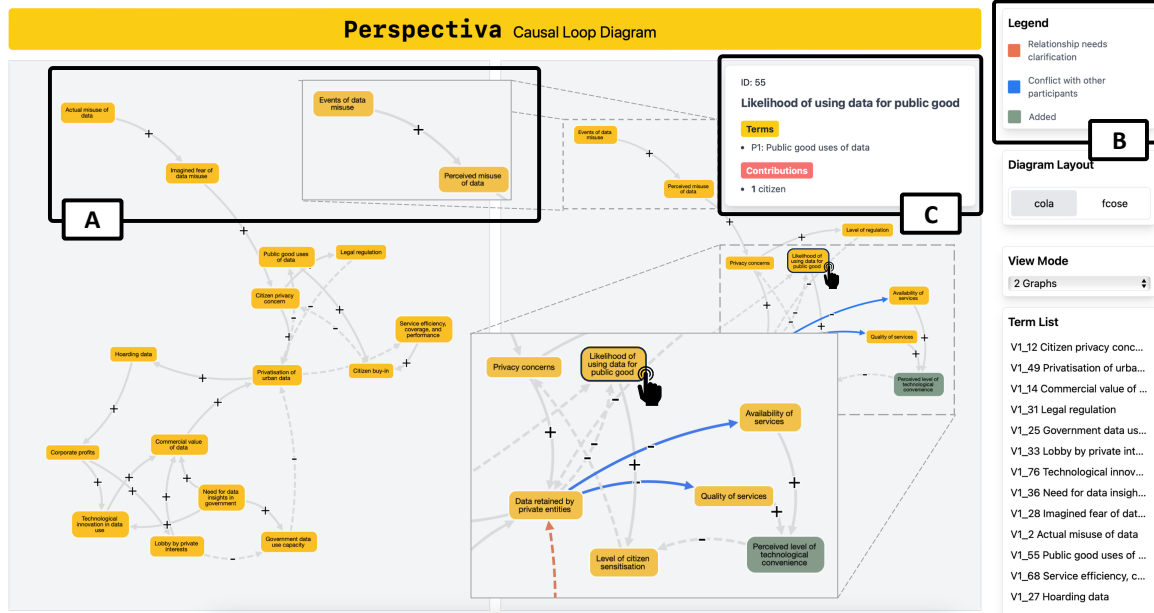


Figure 1: Illustrating *Perspectiva*, a tool that supports C-CLD documentation and transparency via: (A) side-by-side navigation with anchored nodes to facilitate comparison; (B) colour-coding of changes, conflicts, and clarifications; and (C) hover pop-ups that preserve term provenance and identify contributing stakeholder groups.

ABSTRACT

Collaborative causal loop diagrams (C-CLDs) help decision makers to model complex systems and processes, but existing tools offer little support for documenting the model-building process or capturing the provenance of stakeholder contributions. In this paper, we map the C-CLD design space to derive concrete requirements for documentation and transparency. We introduce *Perspectiva*, an interactive prototype shaped by those requirements and refined through iterative feedback. *Perspectiva* enables side-by-side navigation and comparison of CLDs, codifies changes and conflicting relationships, and preserves term provenance and contributor attri-

bution. Its core features include anchored nodes for topological consistency, node interaction, hover-activated provenance pop-ups, and colour-coded encodings. In user studies with domain and visualisation experts, participants reported that *Perspectiva* improved navigation, comparison, and provenance tracking relative to static diagrams as well as highlighting opportunities for enhancement and future research.

Index Terms: Causal loop diagram, Collaborative, Conceptual Modelling, Systems thinking, Documentation

1 INTRODUCTION

Wicked problems, such as climate change or the housing crisis, are made complex by interdependencies between various components and by stakeholders whose priorities often diverge or even conflict. Effective decision-making in such complex systems depends on clearly representing and communicating the interactions among system components and the different stakeholder perspectives that shape the articulation of those interactions. This requirement for clarity emerges not only within organisations (e.g., businesses) but also among groups and communities, such as those pursuing urban sustainability [9], tackling anthropogenic pollution [4], or advancing people-centred data-governance in cities, as illustrated in this paper (Sec. 2).

Conceptual models of such systems draw from graph (network)

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structures and are often represented by Causal Loop Diagrams (CLDs) (Fig. 2). CLDs have been introduced as part of a *systems thinking* movement [16]. They are schematic representations that consist of causal links between variables (Sec. 3.1). CLDs accommodate multi-causality and non-linearity, making them particularly suitable for understanding complex systems. In particular, collaborative model-building using CLDs—referred to as Collaborative Causal Loop Diagrams (C-CLDs) in this paper—is considered a powerful tool for understanding wicked problems that enmesh multiple stakeholders with incongruous perspectives [26].

With the shift in C-CLD usage from consensus-building [23] (e.g., establishing a unified understanding of a problem among stakeholders) to foregrounding multiplicity by integrating heterogeneous, minority, and marginalised perspectives in decision making [14, 13, 4], the need arises for documentation methods that ensure all viewpoints are preserved without being unintentionally altered by researchers or overridden by more influential stakeholders. However, few software tools support the development and manipulation of CLDs, and those available lack features for preserving provenance and documenting key decisions. This drives researchers to rely on labour-intensive workarounds, such as saving successive static screenshots [10] or manually colour coding changes [19]. These practices are time-consuming, error-prone, and inadequate for capturing the step-by-step evolution of a model. As a result, researchers lack practical tools to audit, reproduce, or document the evolution of C-CLDs.

This paper aims to develop computational support for C-CLDs. Although the process of developing C-CLDs calls for tools inherently capable of documenting and depicting the evolution of diagrams, it has not yet been examined by the visualisation community. Our goal is to make the development of C-CLDs transparent and reproducible by capturing the provenance of participants' perspectives alongside researchers' decisions, assumptions, and modifications. To this end, we contribute: (1) a mapping of critical points in C-CLD evolution to opportunities for computational support for provenance documentation and transparency (Sec. 4); (2) translation of those observations into concrete design requirements (Sec. 5); (3) a preliminary prototype, *Perspectiva*, grounded in those requirements and refined through iterative feedback (Sec. 6); (4) findings from exploratory and formative user studies (Sec. 7); and (5) reflection on the contributions and limitations of our approach while outlining future directions for this emerging design space and for *Perspectiva* itself (Sec. 9).

2 CONTEXT

This work is primarily motivated by the first author's firsthand experience with the inherent shortcomings of current tools for documenting C-CLD evolutions and contributions. It emerged from a year-long participatory modelling activity that aimed to co-design a conceptual model of data governance in Melbourne, Australia [3]. The overarching goal of the C-CLD process was not to achieve consensus but to embrace diverse perspectives on data governance issues and foster agonism (i.e., treat conflicting viewpoints as constructive). However, these aims introduced technical challenges in documenting, preserving, and tracking evolving viewpoints and researcher-led assumptions or modifications. Such challenges stemmed largely from the absence of suitable tools that enable seamless documentation of the process. This use case presented the first opportunity to identify the requirements for documenting C-CLD processes and explore and evaluate a working prototype with participants (see Sec. 7).

3 BACKGROUND

3.1 Collaborative Causal Loop Diagrams (C-CLDs)

In its basic form, a CLD comprises variables (nodes) (i.e., elements that can increase or decrease) connected by causal links. The signs

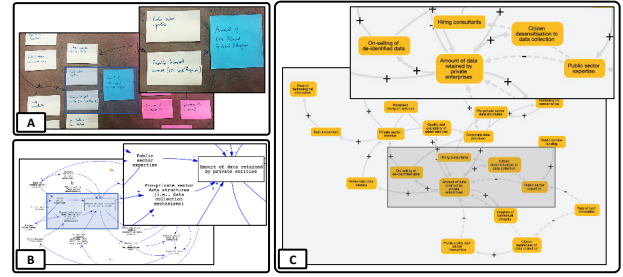


Figure 2: Illustrating Casual Loop Diagrams (CLDs): (A) An individual CLD constructed by a participant using pen and sticky notes, (B) Digitised, Static version of A using Vensim software, (C) Digital, interactive prototype of A *Perspectiva*

of these links denote the type of relationship between variables: a '+' indicates a proportional relationship, while a '-' signifies an inversely proportional relationship (Fig. 1) [16]. Collaborative approaches to developing CLDs vary (e.g., Group Model Building [23], Community-Based System Dynamics [11], and Participatory Causal Loop Diagrams [14]). Although these methods differ in purpose and technique, they all involve multiple participants in constructing the CLD. In this research, all collaborative processes for developing CLDs are referred to as 'Collaborative Causal Loop Diagrams' (C-CLDs). A thorough review of the literature reveals that C-CLDs generally fall into one of two categories: those that include a merging subprocess (e.g., [13, 4]) and those that do not (e.g., [23, 11]). This constitutes the only significant distinction between the methods considered for this study.

3.2 Documentation and Visualisation of C-CLDs

The documentation of changes during the C-CLD development process is seldom mentioned in the literature, despite prominent researchers highlighting the need for better documentation practices [24, 12]. To our knowledge, little work has addressed this issue. Common C-CLD authoring tools, such as Vensim¹ (see Fig. 2 B) and Stella² [19, 13, 5], were primarily developed for system dynamics modelling. While they can capture complex models, they offer limited support for the interactive exploration needed to integrate and understand how different perspectives contribute to a comprehensive model of system actors, relationships, and processes. Similar challenges exist with brainstorming and flowchart applications, such as MiroTM or CanvaTM, which support collaborative diagramming but lack inherent and dynamic representations of contribution provenance, and therefore do not capture how perspectives differ or intertwine. Additionally, Gephi, commonly used for network analysis and visualisation, has been employed to support the C-CLD development process [10]. However, Gephi is not specifically designed for constructing quantitative or conceptual systems models. For instance, users cannot assign polarities to arrows. Moreover, Gephi lacks the incremental graph editing capabilities provided by Vensim and Stella. Finally, the visualisation of causal networks has also been explored [27]; nevertheless, this work does not address the specific requirements of C-CLDs, particularly the preservation of contribution provenance across differing perspectives.

As a result, some researchers have adopted alternative methods to document parts of the C-CLD evolution process. These include saving different C-CLD versions [10], colour coding modifications from workshops [19], using symbols to indicate conflicting relationships [13], and tabulating exclusive stakeholder contributions [4]. While these approaches are useful, they must be manually

¹<https://vensim.com/>

²<https://www.iseesystems.com/store/products/stella-online.aspx>

created, which risks missing details and does not guarantee reproducibility or transparency.

CLDs are often visualised using tools like Vensim, Stella, or similar applications [2, 25, 8]. A C-CLD may be presented as a complete model [22, 17] or as sub-modules [21, 5]. However, existing C-CLD visuals are often static, with documentation limited to ad-hoc colour coding or symbolic annotations. As a result, they cannot capture the step-by-step evolution of a diagram, record provenance, or make conflicting stakeholder perspectives explicit. Our research therefore aims to address these shortcomings that currently leave researchers without any practical means to audit, reproduce, or transparently communicate C-CLD development.

4 POINTS OF EVOLUTION OF C-CLDs

We outline the points of evolution of C-CLDs (E1-E5), i.e., the stages at which CLDs are modified during their development.

E1. Verification of CLDs: After constructing a preliminary CLD the modelling team prepares it for validation. This involves adding variables to fill gaps and enhance logical flow [21], comparing the CLD against recordings, transcripts, or notes to identify and add missing variables and links [5, 6, 11, 15], and aligning terminology to ensure that: (1) variables are depicted as capable of increasing or decreasing [13, 4], and (2) neutrality is by removing quantifiers, adjectives, and adverbs [5, 13].

E2. Standardisation of terms: If the C-CLD development process includes a merging step, terms from different CLDs are standardised to facilitate integration, ensuring that the original meaning is preserved [5, 10]. For example, if one participant uses “data collected by a private entity” and another uses “data collection by private enterprises”, both terms may be standardised to “amount of data collected by private entities”.

E3. Validation: Changes made by researchers after the verification and standardisation of individual CLDs are typically validated [22, 28]. Additionally, the merged C-CLD is generally validated [5, 13]. Validation may involve participants (individually or collectively in workshop sessions) [13, 15, 18, 28], or the model may be cross-checked against transcripts and recordings [13, 5].

E4. Conflict resolution: Conflicts arising during the C-CLD development process become critical during the merging stage when different participants present explicitly conflicting relationships. Such conflicts include disagreements on whether relationships are proportional or inversely proportional, variations in detail, or differences in directions of causality [13]. Researchers have addressed these issues by consulting recordings and transcripts to ensure consistent variable meanings [11, 10], and by using workshops [13] or discussions with stakeholder groups [4] for resolution.

E5. Merging: The merging subprocess involves combining multiple CLDs. Approaches vary: some merge CLDs incrementally [5, 14], others start with the most comprehensive model [13]. During this process, the assignment of participant contributions is often lost and effort is required to preserve it [4].

5 VISUALISATION DESIGN REQUIREMENTS

Following our investigation into the points of evolution of C-CLDs, we derived design requirements for a prototype that facilitates the documentation and visualisation of provenance.

R1. Support navigation between individual CLDs: Navigating between the original and verified CLDs is essential to depict the evolution of CLDs, allowing participants and researchers to review the original work and its modifications [6, 15, 18]. R1 supports E3.

R2. Support the depiction of changes and conflicts: In addition to navigation, the ability to trace and track changes is crucial. This includes allowing researchers to clearly depict modifications, such as the addition, alteration, or removal of variables and relationships, and allowing participants and researchers to easily identify, trace, and preserve the provenance of these changes. The depiction

of conflicting relationships would also be desirable for the conflict resolution stage [13]. R2 supports E1-4.

R3. Document the provenance of terminology: To enhance transparency, it is essential to document the provenance of the terminology used and how it has evolved throughout the CLD iterations. This documentation also supports the validation stage, enabling participants to review how terms have changed in their own CLDs or, in the C-CLD, to examine the various terms that underpin the chosen terminology. R3 supports E2-3.

R4. Document participant contributions in the merged CLD: Documenting how different participants contribute to various nodes during the merging process supports both the validation phase and the subsequent analysis. This documentation is particularly valuable when multiple stakeholders are involved in a challenge, as it highlights diverse perspectives. To preserve anonymity while fulfilling this purpose, contributions should be assigned based on participant groups relevant to the challenge (e.g., in data governance, groups might include data advocates and citizens). R4 supports E5.

6 PERSPECTIVA: IMPLEMENTATION AND DESCRIPTION

Our prototype, *Perspectiva*, aims to fulfil the design requirements outlined in Sec. 5. The development is a work in progress, with iterative adjustments made based on continuous feedback.

Implementation: *Perspectiva*'s user interface was built with Next.js³ and Cytoscape [20]. *Perspectiva* extends the network visualisation capabilities of Cytoscape adding support for side-by-side CLD comparisons with synchronisation of shared-node positions in each view. To guarantee that all shared nodes appear in exactly the same locations, we capture each node's computed (x, y) position from the original version's layout using a one-time 'layoutstop' handler and then reapply those coordinates to the final version via Cytoscape's 'preset' layout. A lightweight diff-and-patch utility updates only the changed elements, which avoids the need for a full re-layout, before forcing the saved positions for every common node. This gives us fixed anchor points for key variables, while still allowing the newly added nodes to be laid out (or manually dragged) around them. The underlying layout algorithms used were based on 'cytoscape-fcose' [1], for initial untangling, and 'cytoscape-cola' [7] to obtain compact but non-overlapping arrangements.

Current Features: *Perspectiva* provides several key features to enhance documentation and the portrayal of changes in C-CLD development processes (see the provided video). Currently, these features serve as a basis for user feedback, and the interactions and responses gathered will guide our final design decisions. Although not directly tied to specific design requirements, one such feature addresses difficulties faced by participants in determining whether links are positive or negative. *Perspectiva* uses a node-link diagram with dual encodings: plus and minus signs, coupled with solid (positive) and dashed (negative) line styles (Fig. 1). The signs appear mid-link rather than above it (as in Vensim), simplifying the depiction of multiple links connecting to the same variable.

Perspectiva addresses R1 in three key ways. First, it presents CLDs in a side-by-side layout that enables seamless navigation and comparison between diagrams (Fig. 1). Second, it anchors the original nodes (i.e., those contributed by participants, even if later modified) in fixed positions across both diagrams, while newly added nodes are arranged around them, ensuring topological consistency between diagrams (Fig. 1 A). Additionally, *Perspectiva* allows users to manually reposition nodes to improve clarity or emphasise specific contributions, and it offers two alternative layout algorithms for exploring different spatial arrangements. To meet R2, *Perspectiva* visually encodes added, conflicting, and ambiguous nodes and links using colour coding, accompanied by a corresponding legend (Fig. 1 B). In support of R3, it provides interactive node inspection:

³<https://nextjs.org/docs>

Table 1: Mean expert feedback scores (out of 5).

	Navigation	Comparison	Colour codes	Term changes
Interactive	4.5 	4.0 	4.6 	3.6 
Static	2.9 	3.5 	3.9 	3.3 

when users hover over a node, a pop-up displays the original term provided by participants (for nodes whose labels have been modified) (Fig. 1 C). This pop-up also indicates the participant group (e.g., advocate, public sector), which will later enhance the depiction of contributions in the merged C-CLD, thereby addressing **R4**.

7 FORMATIVE USER STUDY

Description. The use case described in Sec. 2 served as a testbed for trialling static CLD representations and an earlier version of *Perspectiva*. Eight participants constructed their own CLDs in individual sessions (4 using paper and sticky notes (Fig. 2A), 4 using Miro). Two additional CLDs were transcribed from interviews. These CLDs were then verified, standardised, validated, merged, and used as scoping tools in a subsequent workshop. During the validation stage, an older version of *Perspectiva* was tested alongside static representations: half the participants received only static versions of their original and modified CLDs, while the other half interacted with an early version of *Perspectiva* that lacked side-by-side navigation and anchored nodes. Feedback on both static and interactive representations guided the development of this version.

Findings. Unsolicited participant feedback revealed difficulties with the static version, regarding (1) navigation between original and modified versions, (2) comparing or matching nodes across versions, and (3) associating polarities with links. One participant who received the older *Perspectiva* prototype also reported navigation challenges, primarily due to the absence of a side-by-side view and changes in node positions between the original and modified CLDs.

8 QUALITATIVE USER STUDY

Description. A user study was conducted to inform iterative improvements. It involved comparing static and interactive representations of C-CLDs and investigating *Perspectiva*. Two C-CLD experts and five visualisation experts were included. Each participant received a user guide containing: (1) clarifications of key terminology used in the questionnaire, (2) instructions for completing the questionnaire, (3) a link to *Perspectiva* with usage guidelines, (4) static versions of the original and modified CLDs, and (5) a link to the questionnaire. While the C-CLD experts completed the questionnaire independently, the first author guided the visualisation experts in person or via Zoom. Participants were encouraged to ask questions about the tool and to give feedback. Tasks alternated between static and interactive CLD versions. There were no restrictions on completion time, but the average was 30 minutes.

Findings. Findings from both the static and interactive versions were organised into four dimensions: navigation, comparison, colour-coded changes, and term changes. In every dimension, the mean user rating was higher for the interactive version than for the static version (see Table 1). *Navigation*: Three experts noted that the side-by-side layout facilitated navigation. One visualisation expert highlighted the value of having two layout algorithms to explore. *Comparison*: Two experts observed that the topological consistency between the original and modified diagrams facilitated the comparison process. One C-CLD expert commented that manual node repositioning allowed for more focused comparisons. Two visualisation experts further recommended adding filters to isolate specific nodes and their immediate connections, enabling users to concentrate on particular sections of the diagram. One visualisation expert noted that modifications to node labels complicated direct comparison. *Colour-coded changes*: A C-CLD expert noted that

colour codes may present challenges for users with colour-vision deficiencies. One C-CLD expert suggested that the legend should “indicate ‘Variables added by researcher’, to make it clear that these are not variables added by participants.” They also recommended the depictions of changes in links. *Term changes*: Two visualisation experts reported difficulty matching nodes whose labels had changed across versions with precision, while a C-CLD expert found that “hovering over the variables [...] easily [shows] original wording.” Two participants suggested a ‘track changes’ feature on node labels to explicitly mark wording edits, and three participants recommended linking corresponding nodes so that selecting a node in one version highlights its counterpart in the other. Although we focused on individual CLD validation rather than examining contribution assignment within group C-CLDs, one participant suggested that the hover pop-up display the number of contributors to each node or link. They also recommended visually encoding contribution frequency by varying edge weights and node sizes according to the number of contributions.

9 DISCUSSION AND CONCLUSION

Side-by-side navigation and topological consistency effectively supported **R1**. However, exploration tools, such as filters, are needed for targeted comparisons. Because node labels often changed during the standardisation phase, direct comparison sometimes remained time-consuming. For **R2**, the legend should be modified to distinguish researcher-added from participant-added elements. Additionally, shapes should be used to improve accessibility for those with colour-vision deficiencies. **R3** proved challenging: hover pop-ups alone did not reliably reveal node correspondences. To address this (1) a ‘track changes’ feature on labels should be added to mark edits explicitly, and (2) matching nodes should be linked so that selecting one highlights its counterpart. Finally, for **R4**, rather than withholding contribution counts until after merging, participants should be able to see how many stakeholders contributed to each node during individual validation. Our user study highlighted distinct but complementary roles domain and visualisation experts play in advancing C-CLD tooling. C-CLD experts focused on the accuracy of CLD content, while visualisation experts proposed features that draw on the affordances of visualisation systems, which were less familiar to domain experts.

This work has several limitations. First, it presents a preliminary prototype, situating its contribution in mapping a design space for computational support in documenting C-CLD evolutions. By focusing primarily on documentation, we acknowledge that we have not addressed the need for computational support in other C-CLD processes. For example, the **merging** of C-CLDs remains a manual, labour-intensive task [13], and C-CLD **analysis** still relies on manually arranging and identifying reinforcing and balancing loops without interactive tooling [4]. Additionally, how to engage multiple participants in understanding and providing feedback on C-CLDs during workshops remains unexplored. The user study was also limited to only two C-CLD experts as it is a niche participant group. We hope that further work to facilitate the documentation and transparency of C-CLDs will enable the method to be a more common choice. Finally, although C-CLDs are structured diagrams that follow specific conventions, they still involve multiple iterations and changes in media (e.g., from paper to digital) and in structure. Hence, the examination of how the rigidity of a tool like *Perspectiva* impacts user validation and proposed changes compared to more sketchy media is needed to ensure participants’ perspectives are not compromised [29]. Future work includes: developing the features outlined earlier in this section, conducting a formal evaluation, developing and evaluating the documentation of the merged C-CLD, and mapping the design space that characterises computational needs for merging, analysing, and facilitating the use of C-CLDs in workshops.

ACKNOWLEDGMENTS

We thank the domain and visualisation experts for taking time out of their busy schedules to participate in this study.

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